Software Requirements Specification (SRS)

Main Project: Airline Customer Satisfaction & Service Quality Analysis — Analyzing passenger Experience and Service Quality

1. Introduction and Purpose

1.1 Project Background

Airline services play an important role in passenger satisfaction. Passenger ratings provide information about their overall experience and different aspects of airline services, such as seat comfort, cabin staff service, food and beverages, and Wi-Fi connectivity. Analyzing these ratings can help identify service areas that may need improvement.

1.2 Purpose

The purpose of this project is to analyze airline passenger ratings to understand customer satisfaction and service quality. The project aims to compare ratings across service categories, airlines, traveller types, and seat types. It will also examine the relationship between overall ratings and whether passengers recommend the airline.

1.3 Stakeholders

The main stakeholders who may benefit from this analysis are:

- Airline management: To understand passenger satisfaction and identify areas for possible service improvement.
- Customer service teams: To understand passenger feedback about different services.
- Service quality teams: To compare service ratings and identify lower-rated areas.

1.4 Intended Users

The intended users of this project are airline analysts, customer experience teams, service quality teams, and anyone interested in understanding airline passenger satisfaction through data analysis.

2. Problem Definition and Scope

2.1 Business Problem

Airlines need to understand passenger satisfaction and identify service areas that may require improvement. This project analyzes existing passenger review data to understand overall satisfaction,

compare ratings across seven service categories, and examine differences in ratings across airlines and passenger groups. The findings will help identify patterns in customer feedback and support data-driven decisions to improve service quality.

2.2 Project Objectives

- Classify overall passenger ratings into Poor (1–2), Medium (3–4), and Positive (5–10) categories, and calculate the percentage of ratings in each category**.**
- Calculate and compare average ratings across seven service categories.
- Identify the three lowest-rated service areas.
- Compare average overall ratings between passengers who recommend the airline and those who do not.
- Compare average ratings across airlines, traveller types, and seat types.
- Present key findings using statistical summaries, charts, and visualizations.

2.3 Inclusions

The project includes:

- Loading and inspecting the passenger review dataset using Python and Pandas.
- Checking and handling missing values, duplicate records, inconsistent text values, and incorrect data types where necessary.
- Performing data cleaning and feature engineering.
- Conducting univariate, bivariate, and multivariate exploratory data analysis (EDA).
- Calculating descriptive statistics, average ratings, and satisfaction percentages.
- Comparing service ratings and passenger satisfaction across relevant categories.
- Creating visualizations using appropriate Python libraries.
- Summarizing insights and identifying potential service improvement areas.

2.4 Exclusions

The project does not include:

- Developing a software application, website, or mobile application.
- Building a live airline customer feedback system.
- Predicting future satisfaction using machine learning models.
- Analyzing verified flight delays, cancellations, or operational performance using separate flight records.
- Collecting new passenger reviews or conducting surveys.
- Verifying the authenticity of individual reviews.
- Calculating actual financial costs or measuring the impact of service improvements.

3. Data Requirements

3.1 Dataset Source

The dataset used for this project contains airline passenger reviews and ratings obtained from the Mendeley Data repository.

- Dataset Source: Mendeley Data –Airline Travel Reviews Data(Skytrax)
- Dataset Link: https://data.mendeley.com/datasets/vgjk58vf8h/1

3.2 Dataset Structure

- The dataset contains 64,740 rows and 19 columns, based on the initial dataset inspection.
- Each row represents a passenger review of an airline and contains information about the passenger's experience, service ratings, and recommendation.
- The dataset includes numerical and categorical columns.

3.3 Key Fields Used in the Analysis:

- over_all_rating : Overall passenger rating (1–10)
- airline_name: Name of the airline being reviewed
- seat_comfort: Seat comfort rating (1–5)
- cabin_staff_service: Cabin staff service rating (1–5)
- food__beverages: Food and beverages rating (1–5)
- inflight_entertainment: Inflight entertainment rating (1–5)
- ground_service: Ground service rating (1–5)
- wifi__connectivity: Wi-Fi and connectivity rating (1–5)
- value_for_money: Value for money rating (1–5)
- recommended: Whether the passenger recommends the airline
- type_of_traveller: Category of traveller
- seat_type: Type of seat or travel class
- review_date: Date the review was submitted
- trip_verified: Indicates whether the trip is verified

  Rating Scales: The over_all_rating field uses a 1–10 scale, while the seven individual service rating fields use a 1–5 scale.

3.4 Minimum Quality Needs

- Identify and appropriately handle missing values.
- Check for duplicate records.
- Validate overall ratings against the 1–10 scale and service ratings against the 1–5 scale.
- Convert review dates to the appropriate date format.
- Standardize inconsistent text values where necessary.

- Check and correct data types where required.

3.5 Data Access

The dataset was downloaded from the Mendeley Data repository in Excel format (.xlsx) and converted into CSV format (.csv) for analysis. The CSV file is stored locally and loaded into a Jupyter Notebook using Python and Pandas.

4. Functional Requirements

4.1 Data Loading: Load the airline passenger review dataset from a CSV file using Python and Pandas.

4.2 Data Preprocessing: Identify and handle missing values, check duplicate records, correct data types, and standardize inconsistent values where necessary.

4.3 Data Analysis: Perform univariate, bivariate, and multivariate analysis to identify passenger satisfaction patterns and evaluate airline service quality.

4.4 KPI Calculations: Calculate satisfaction-category percentages, average ratings for the seven service categories, and identify the three lowest-rated service areas. Compare overall ratings across airlines, traveller types, seat types, and recommendation status.

4.5 Data Visualization: Create suitable charts and graphs to present rating distributions, service quality comparisons, and customer satisfaction patterns.

4.6 Reporting: Summarize the key findings and insights to identify service areas that may require improvement.

5. Non-Functional Requirements

- Tools and Environment: Python, Jupyter Notebook, Pandas, and visualization libraries.
- Reproducibility: The notebook should produce consistent results when run with the same dataset and settings.
- Documentation: Include clear headings, code explanations, and summaries of findings.
- Performance: The analysis should run within a reasonable time.
- Reliability: The notebook should run without errors, and calculations should be accurate.

6. Expected Outputs and Acceptance Criteria

6.1 Expected Outputs

- A cleaned airline passenger review dataset.
- A Jupyter Notebook containing data preprocessing, analysis, and visualizations.
- Calculated satisfaction percentages and average ratings for seven service categories.

- Identification of the three lowest-rated service areas.
- Charts, key findings, and recommendations for potential service improvements.

6.2 Acceptance Criteria

The project will be considered complete when:

- The notebook runs from beginning to end without errors.
- Data cleaning and analysis are completed successfully.
- Each project objective is addressed using relevant metrics or charts.
- The three lowest-rated service categories are identified correctly.
- Visualizations and findings are clearly presented and consistent with the analysis results.

7. Assumptions, Constraints & Dependencies

- Assumptions: Passenger ratings represent customer feedback.
- Constraints: Missing values and incomplete reviews may affect the analysis.
- Dependencies: The dataset, Python, Pandas, and required visualization libraries must be available.
- Approvals: Dataset selection and project scope are subject to trainer approval.